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# INVENTIVE DISCLOSURE - CONFIDENTIAL

| **Title of the Invention** |
|---|
| Smart Exhaust Fan using STM32 bluepill board |
| **What is the Technical Field of Invention?**<br>*(Should include Problem and Solution statement and general purpose in relation with your invention)* |

. The present invention relates to the field of **embedded systems, smart home automation, and environmental monitoring systems**.

## Problem Statement:

Conventional exhaust fans operate manually or on simple timers without considering environmental conditions such as air quality, humidity, temperature, or gas concentration. This leads to:

- Inefficient ventilation
- Excess energy consumption
- Poor indoor air quality

## Solution Statement:

The proposed invention introduces a **smart exhaust fan system** using an STM32 Bluepill microcontroller that:

- Automatically monitors environmental parameters
- Activates/deactivates the fan intelligently
- Optimizes energy consumption
- Improves indoor air quality

## General Purpose:

To create an intelligent, automated ventilation system for homes, kitchens, bathrooms, and industrial

environments.

**Search Terms**

*(Please provide key words and common industry phrases on the basis of which this invention could be connected with the prior art)*

- Smart exhaust fan
- STM32 Bluepill
- Air quality monitoring system
- IoT ventilation control
- Automated fan controller
- Gas sensor-based exhaust system
- Embedded smart home devices
- Humidity-based fan control

Background of the Invention (Present state of Art)

## Existing Technologies:

- Manual exhaust fans
- Timer-based exhaust systems
- Humidity-controlled fans
- IoT-based smart fans (limited adoption)

## Limitations:

- Lack of multi-parameter sensing
- No real-time adaptive control
- High energy wastage
- Limited intelligence and automation
- Expensive IoT-based systems

a. What are the present technologies that exist in the field of your invention

Existing systems include:

- Humidity-sensor-based bathroom fans
- Gas detector-triggered ventilation systems
- Arduino-based fan automation projects

b. What are the limitations of the same?

## Limitations of Prior Art:

- Single parameter monitoring
- Low processing efficiency
- Lack of scalability
- No predictive or adaptive control

**Prior Art**

*(Please provide the details of the product or process by the help of which the problem solved by the invention was tackled before this invention. Please details of prior technologies, patents, or publications relevant to this*

*invention)*

- Integrating **multi-sensor data (gas, humidity, temperature)**
- Using a **high-performance STM32 microcontroller**
- Providing **real-time intelligent control**
- Enabling **energy-efficient operation**

**Has the work filled a major gap in the prior art? If yes, a brief description of this gap.**

Yes, the invention fills a major gap by:

- Integrating **multi-sensor data (gas, humidity, temperature)**
- Using a **high-performance STM32 microcontroller**
- Providing **real-time intelligent control**
- Enabling **energy-efficient operation**

**What are the components of the Invention?**
*(Explain the external and internal parts/blocks/systems-subsystems/ composition/steps of process of your invention).*

- STM32 Bluepill microcontroller
- Gas sensor (e.g., MQ series)
- Temperature sensor (e.g., LM35/DHT11)
- Humidity sensor
- Relay module
- Exhaust fan
- Power supply unit
- Optional: LCD/LED display

**How does the Invention work?**
*(This portion of your invention disclosure deals with the working of your invention, the processes involved and the interactive actions taking place between the various components of your invention to give the desired output.)*

- Sensors continuously monitor environmental parameters.
- Data is sent to STM32 Bluepill.
- The microcontroller processes inputs using predefined thresholds.
- If pollution/humidity exceeds limits:
- Relay is triggered
- Exhaust fan turns ON
- When conditions normalize:
- Fan automatically turns OFF
- Optional display shows real-time data

**Novelty of the Invention/Inventive Step of the Invention**
*(Please state the unique/ novel aspect of your invention and how it is different and/or better than the existing technologies):*

- Use of **STM32 Bluepill** instead of conventional controllers
- **Multi-parameter sensing system**
- **Real-time adaptive decision-making algorithm**
- **Energy-efficient automation**

| | |
|---|---|
| 10. | **What are the advantages and improvements over the existing methods/devices? (***Please list advantages and improvements against closest patent and non-patent documents/products in pipeline explaining technical advancement of present invention over the existing technologies***)** |
| | • Reduced power consumption<br>• Improved air quality<br>• Fully automated system<br>• Low-cost implementation<br>• Scalable and customizable<br>• Faster processing compared to Arduino systems |
| 11. | **Do you have in your mind as a researcher/inventor some ideas as to what are the various other/alternative forms in which your Invention can be depicted or utilized, in addition to one basic embodiment that has been described above?**<br>*(This will lay the foundation, for making your patent as broad as possible. These are called preferred embodiments or variants.)* |
| | • IoT-enabled version (mobile app control)<br>• AI-based predictive ventilation system<br>• Solar-powered exhaust system<br>• Integration with smart home ecosystems<br>• Wireless sensor network version |
| 12. | **Please provide final CAD images/line diagrams of the final prototype which illustrates all the essential elements of the invention.**<br>*(Please prepare images in two sets, one completely labeled to details and the second one without any description, legends/labeling or numbering. Include relevant Flow charts.)* |
| | |
| 13. | **Brief description from the researcher/inventor in his own words explaining the images/ drawings.** |
| | |
| 14. | **Please give complete description including the working examples/ working methodology of the invention.** |
| | • In a kitchen environment, when gas concentration increases due to cooking:<br>  ○ MQ sensor detects gas<br>  ○ STM32 processes signal<br>  ○ Fan turns ON automatically<br>• When air quality improves:<br>  ○ System turns OFF fan |
| 15. | **Please provide details of experiments conducted /third party validation data if any.** |
| | • Tested in enclosed room environment<br>• Response time: ~1–2 seconds<br>• Power savings observed vs manual operation<br>• Reliable sensor-triggered operation |
| 1 | **Please provide details of any public disclosure done/ any publication made.** |

**16.**

**17. What is the stage/level of development of the Invention?**

- Prototype developed
- Results validated

**18. What are the aspects of your disclosure that you want to claim/monopolize? (Proposed Claims)**

- A smart exhaust fan system using STM32 microcontroller.
- Multi-sensor based automated ventilation control.
- Real-time environmental monitoring and adaptive fan control.
- Energy-efficient exhaust system with automated switching.

**19. Have you conducted novelty/inventiveness search for your invention? If yes, what are the databases /references used by you? What are the search results?**

| Sno | Reference | Existing Idea | Our Invention |
|---|---|---|---|
| **Sno** | **Reference** | **Existing Idea** | **Our Invention** |
| 1 | Arduino fan systems | Basic automation | Advanced STM32 system |
| 2 | Humidity fans | Single parameter | Multi-parameter sensing |
| 3 | IoT fans | Expensive | Low-cost efficient system |
| | | | |

**20. Do you feel that a person of "average" skill (not-extraordinary skill) in your area of technology would have arrived at your invention with existing knowledge in public domain? If no, what could be the reasons for the same?**

No, an average skilled person would not easily arrive at this invention because:

- Integration of multiple sensors with STM32 is complex
- Requires optimized embedded programming
- Efficient real-time decision-making logic

**21. Kindly provide broad workable ranges for all the parameters involved in your invention.**

# Parameter Ranges

- Temperature: 20°C – 60°C
- Humidity: 30% – 90%
- Gas concentration: Based on sensor calibration
- Voltage: 5V – 12V

**22. Commercialization Data (at least a minimum of 2 companies must be provided and preferably up to 5)**

a. Kindly give the names and complete addresses of five different companies which might be interested in the commercial manufacture of this technology.

Potential companies:

- Honeywell
- Panasonic
- Havells
- Crompton
- Schneider Electric

a. Kindly also enclose a short marketing profile of the technology highlighting the advantages vis-à-vis the prior-art technologies in this area

The **Smart Exhaust Fan using STM32 Bluepill** represents a next-generation solution in the smart ventilation and home automation market. Unlike conventional exhaust systems that rely on manual operation or single-parameter sensing, this invention integrates **multi-sensor intelligence and real-time adaptive control**, offering a highly efficient and cost-effective solution.

## Key Advantages over Prior Art:

b. **Multi-parameter sensing** (temperature, gas) vs single-sensor systems
c. **Real-time automation** vs manual or timer-based control
d. **Energy-efficient operation** reducing electricity consumption
e. **Low-cost implementation** compared to expensive IoT-based systems
f. **Faster processing** using STM32 architecture vs traditional Arduino-based systems
g. **Scalable design** suitable for both domestic and industrial applications

h. This technology is especially suitable for **smart homes, kitchens, bathrooms, laboratories, and industrial ventilation systems**, making it commercially viable and highly adaptable.

**23. Provide available data or information on market potential/size, adoption trends, industry analysis, and projected growth.**

| Year | Estimated Market Size | Growth Drivers | Notable Existing Devices |
|---|---|---|---|
| | | | |
| 2022 | ~$12 Billion | Basic ventilation demand | Humidity-based exhaust fans |
| 2023 | ~$14 Billion | Smart home adoption | IoT-enabled fans |
| 2025 | ~$18 Billion | Air quality awareness | Smart air purifiers + fans |
| 2030 (Projected) | ~$30 Billion+ | AI & automation integration | Fully automated ventilation systems |

The global smart ventilation and air quality control market is growing rapidly due to:

- Rising awareness of **indoor air pollution**
- Growth of **smart homes and IoT devices**
- Increasing **health and safety regulations**
- Demand for **energy-efficient appliances**

## Key Industry Players:

- Honeywell
- Panasonic
- Havells
- Crompton
- Schneider Electric

**24. Please provide the names of the inventors involved in the Invention**

Amrutham
Poornima
Aishwarya M
Keertana K

**25. User Information**

a. Who all are the potential users of the technology?

- Homeowners
- Apartment residents
- Restaurants and kitchens
- Laboratories
- Industrial facilities
- Smart building developers

b. Age group of the user?

- 18–65 years (general users)
- Facility managers and professionals

c. Expected benefits to the user?

- Improved indoor air quality
- Reduced manual effort
- Energy savings
- Increased safety (gas detection)
- Smart automation convenience

d. Cost advantage when compared to available solutions

- 30–50% cheaper than IoT-based smart ventilation systems
- Lower installation and maintenance cost
- Uses affordable embedded components

**26. What is Technology readiness level of your invention? (Tick the appropriate TRL)**

| Research | | | Development | | | Deployment | | |
|---|---|---|---|---|---|---|---|---|
| **TRL 1** | **TRL 2** | **TRL 3** | **TRL 4** | **TRL 5** | **TRL 6** | **TRL 7** | **TRL 8** | **TRL9** |
| Basic Principles observed | Technology concept formulated | Experimental proof of concept | Technology validated in a lab | Technology validated in a relevant environment (Industrially relevant in case of key enabling technologies) | Technology demonstrated in a relevant environment (Industrially relevant in case of key enabling technologies) | System prototype demonstration in an operational environment | System complete and qualified | Actual system proven in an operational environment (Competitive manufacturing in case of key enabling technologies, or in space) |
| | | | | | | | | |

**27. Any additional notes or remarks.**